The Base Semigroup of Holte's Carries Chain: An Exact Route-A Certificate

Route-A structural certificate HCS-C194

Abstract

Adding n independent base-b digit columns produces a Markov chain on the carry states $\{0, \dots, n-1\}$. We source-lock Holte's all-parameter transition formula, mixed-radix semigroup $P_a P_b = P_{ab}$, common eigenvectors, simple spectrum $1, b^{-1}, \dots, b^{-(n-1)}$, and Eulerian stationary law. The diagonalization closes every power trace, finite determinant, and an exact common-projector convergence expansion; a sourced total-variation estimate is kept separate from that algebraic deduction. An exact rational census checks 72 matrices but is only regression evidence. Ordinary positional addition earns a weak arithmetic relation; prime and composite bases obey the same theorem, so no rational-prime orbit semantics, target divisor, or natural target quantization follows.

Keywords: carries; positional addition; Markov chain; Eulerian numbers; matrix semigroup; Route A.

1 The frozen digit-column dynamics

Fix $n \geq 1$ and $b \geq 2$. From a carry-in $i \in X_n = \{0, \dots, n-1\}$, sample $d_1, \dots, d_n$ independently and uniformly from $\{0, \dots, b-1\}$. One clock step returns the unique j such that

$$jb \leq i + d_1 + \cdots + d_n < (j+1)b.$$

Introducing the output remainder as a slack digit gives Holte's coefficient formula [1, Theorem 1]

$$P_b(i,j) = b^{-n}[x^{(j+1)b-1-i}](1 + x + \cdots + x^{b-1})^{n+1}. \tag{1}$$

The chain is irreducible and aperiodic. The case $n = 1$ is the one-state identity, retained as the exact boundary.

2 Base semigroup and complete spectrum

Write a base-ab digit uniquely as $x + ay$, with $0 \leq x < a$ and $0 \leq y < b$. Applying the carry first to the base-a column and then to the base-b column gives the same output as one base-ab column. Therefore

$$P_a P_b = P_{ab}, \qquad P_b^r = P_{b^r}. \tag{2}$$

Holte's Theorem 3 gives a matrix V_n, independent of b, such that

$$V_n P_b V_n^{-1} = \operatorname{diag}(1, b^{-1}, \dots, b^{-(n-1)}). \tag{3}$$

The eigenvalues are distinct. The first row of V_n consists of Eulerian numbers, hence the stationary probability is

$$\pi_n(j) = \frac{\mathsf{A}(n,j)}{n!}, \qquad 0 \leq j < n.$$

It follows immediately that

$$\operatorname{tr}(P_b^r) = \sum_{k=0}^{n-1} b^{-rk}, \tag{4}$$

$$\det(I - zP_b) = \prod_{k=0}^{n-1} (1 - zb^{-k}), \tag{5}$$

$$\chi_{P_b}(t) = \prod_{k=0}^{n-1} (t - b^{-k}). \tag{6}$$

These are finite Markov-operator identities, not an Artin–Mazur or target determinant.

3 Common projectors and convergence

Let $E_0, \dots, E_{n-1}$ be the spectral projectors determined by the base-independent matrix V_n. They are independent of b, and

$$P_b^r - E_0 = \sum_{k=1}^{n-1} b^{-rk} E_k. \tag{7}$$

Here E_0 maps every row distribution to π_n. Equation (7) is an exact identity, not merely an asymptotic statement; in every fixed matrix norm it gives geometric convergence governed by b^{-r} unless the E_1 component vanishes.

Diaconis and Fulman identify the carries law with a descent marginal of repeated riffle shuffles and analyze convergence [2]. Their Theorem 3.3 supplies, for $n \geq 3$, every start i, and $r \geq 0$,

$$\|P_b^r(i, \cdot) - \pi_n\|_{\mathrm{TV}} \leq \frac{(n-1)/2 + i}{b^r}. \tag{8}$$

We attribute this bound rather than presenting it as a package theorem. For $n = 1$ the chain is already stationary; $n = 2$ follows directly from the two eigenvalues $1, b^{-1}$ and is not folded into the stated $n \geq 3$ source locator.

4 Exact regression certificate

The producer reconstructs digit-sum coefficients by convolution. A separate checker uses slack-variable inclusion–exclusion, enumerates Eulerian numbers from permutations, and obtains characteristic coefficients through the Faddeev–LeVerrier algorithm. SymPy independently checks matrix polynomials, eigenspace dimensions, stationary equations and traces.

The frozen census contains 72 cases ($1 \le n \le 8$, $2 \le b \le 10$), 1,836 transition cells, 392 base-semigroup tuples and 96 power-identity tuples. Bases $2, 3, 5, 7$ contribute 32 prime-tagged cases; $4, 6, 8, 9, 10$ contribute 40 composite-tagged controls. Enumeration tests code only: the all-parameter quantifiers belong to Holte's theorem.

5 Route-A verdict

Positional addition is genuinely arithmetic, but the integer base supplies no rational-prime primitive carrier, prime-power repetition law, or $\log p$ clock. The frozen object is a stochastic matrix rather than a deterministic primitive-orbit map. Its finite determinant has no target divisor, and a self-adjoint similarity chosen after diagonalization would be noncanonical. Thus

$$(A0, A1, A2, A3, A4) = (\mathrm{WEAK\_ARITHMETIC\_RELATION}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FORMAL\_HINT}).$$

The overall verdict is rejection and Route B is false. No target tables, arithmetic local data, Euler factors, root numbers, automorphy, target functional equation, or Hilbert–Pólya operator is claimed.

References

[1] J. M. Holte, Carries, combinatorics, and an amazing matrix, *Amer. Math. Monthly* 104 (1997), 138–149, doi:10.1080/00029890.1997.11990612.

[2] P. Diaconis and J. Fulman, Carries, shuffling, and symmetric functions, *Adv. Appl. Math.* 43 (2009), 176–196, doi:10.1016/j.aam.2009.02.002.